Practice workbook · Finding n from an equation linking sums and terms

Substituting the sum and nth-term formulas into a given equation and solving the quadratic for a positive whole number n.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-06

1. Question 1 · part b · 5 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part b only.

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1 The nth term of an arithmetic series is t_n and the common difference of the series is d.

Given that $t_2 + t_9 = 0$ and that $t_4 + t_6 + t_{10} = 14$

(a) (i) show that $d = 4$

(ii) find the first term of this series.

(4)

A different arithmetic series A has first term 24 and common difference 6
For series A, the sum of the first $2n$ terms is 3 times the sum of the first n terms.

(b) Find the value of n.

(5)

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1. Question 1 · part b · 5 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part b only.

Question 1 continued

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1. Question 1 · part b · 5 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part b only.

Question 1 continued

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(Total for Question 1 is 9 marks)

1. Question 1 · part b · 5 marks · Working and answer

Working and answer

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2. Question 6 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

6 (a) Show that $\sum_{r=1}^{n}(4r-3) = n(2n-1)$

(3)

(b) Hence, or otherwise, find the least value of n such that $\sum_{r=1}^{n}(4r-3) > 1000$

(3)

Given that $S_n = n(2n-1)$, $t_n = (4n-3)$ and that $18 + 3t_{n+7} = S_{n+4}$

(c) find the value of n.

(4)

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2. Question 6 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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2. Question 6 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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2. Question 6 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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(Total for Question 6 is 10 marks)

2. Question 6 · part c · 4 marks · Working and answer

Working and answer

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3. Question 6 · part c · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

6 An arithmetic series A has first term a and common difference d.

The sum S_n of the first n terms of A is given by $S_n = n(15 + 2n)$

(a) Find the value of a and the value of d.

(4)

(b) Find the 20th term of A.

(2)

Given that $S_{2p} - 2S_p = 1 + S_{(p-1)}$

(c) find the value of p.

(4)

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3. Question 6 · part c · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.



3. Question 6 · part c · 4 marks · Working and answer

Working and answer

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4. Question 8 · part b · 4 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

8 The sum to n terms of an arithmetic series A is S_n

The sum of the first four terms of A is 42 and the fifth term of A is 23

(a) Show that $S_n = \sum_{r=1}^{n}(Pr - Q)$ where P and Q are prime numbers.

(6)

$S_{2n} - 3U_n = 1062$ where U_n is the nth term of A

(b) Find the value of n

(4)

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4. Question 8 · part b · 4 marks

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Further practice · Attempt part b only.

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Question 8 continued

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(Total for Question 8 is 10 marks)

4. Question 8 · part b · 4 marks · Working and answer

Working and answer

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5. Question 6 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

6 The sum of the first n terms of an arithmetic series is S_n where $S_n = n(3 + 2n)$

(a) Find the value of S_{20}

(2)

Given that $S_n = \sum_{r=1}^{n} (Ar + B)$

(b) find the value of A and the value of B

(6)

A different arithmetic series has first term 7 and common difference 4

The sum of the first n terms of this series is T_n

(c) Use algebra to find the value of n for which $T_n = S_n + 252$

(5)

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5. Question 6 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

Question 6 continued

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5. Question 6 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

Question 6 continued

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5. Question 6 · part c · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

Question 6 continued

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(Total for Question 6 is 13 marks)

5. Question 6 · part c · 5 marks · Working and answer

Working and answer

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6. Question 7 · part c · 5 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

7 An arithmetic series P has first term a, common difference d and nth term u_n

Given that $u_5 = 4x + 6$ and that $u_8 = 7x + 3$

(a) (i) show that $d = x - 1$

(ii) find the value of a

(4)

Given further that $u_9 = 42$

(b) find the value of x

(2)

The sum of the first n terms of P is S_n

(c) Find the value of n for which $S_{(n+1)} = 12u_n + 18$

(5)

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6. Question 7 · part c · 5 marks

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Further practice · Attempt part c only.

Question 7 continued

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6. Question 7 · part c · 5 marks

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Further practice · Attempt part c only.

Question 7 continued

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6. Question 7 · part c · 5 marks

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Further practice · Attempt part c only.



Question 7 continued

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(Total for Question 7 is 11 marks)

6. Question 7 · part c · 5 marks · Working and answer

Working and answer

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7. Question 9 · part c · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

9 (a) Show that $\sum_{r=1}^{n}(5r-1)=\frac{n}{2}(3+5n)$ (3)

(b) Hence, or otherwise, evaluate $\sum_{r=10}^{20}(5r-1)$ (3)

The sum of the first n terms of an arithmetic series is S_n where $S_n=\sum_{r=1}^{n}(5r-1)$

The rth term of this series is u_r

Given that $S_n=12u_{n+1}+52$

(c) find the value of n (5)

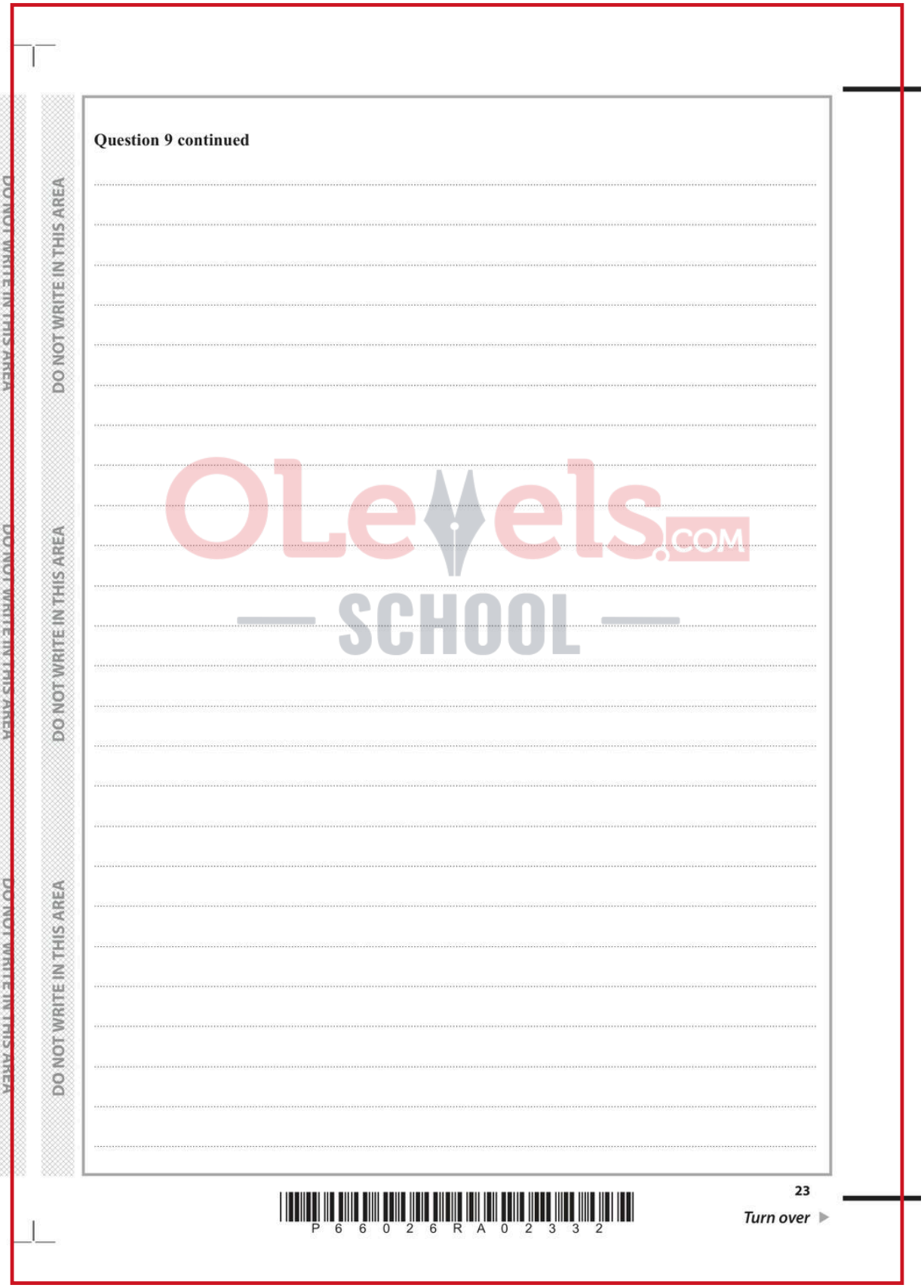
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7. Question 9 · part c · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.



Question 9 continued

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7. Question 9 · part c · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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7. Question 9 · part c · 5 marks

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Question 9 continued

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(Total for Question 9 is 11 marks)

7. Question 9 · part c · 5 marks · Working and answer

Working and answer

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8. Question 2 · part c · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part c only.

2 The sum of the fifth, sixth and seventh terms of an arithmetic series A is nine times the sum of the first and second terms.
The third term of A is 12

(a) Find the first term and common difference of A

(5)

The nth term of A is u_n

(b) Find the value of $\sum_{r=15}^{60} u_r$

(4)

The sum to n terms of A is S_n

Given that $2S_n - 5u_n = 10$

(c) find the value of n

(4)

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8. Question 2 · part c · 4 marks

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Question 2 continued

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(Total for Question 2 is 13 marks)

8. Question 2 · part c · 4 marks · Working and answer

Working and answer

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9. Question 5 · part c · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

5 (a) Show that $\sum_{r=1}^{n}(3r+5) = \frac{1}{2}n(3n+13)$ (3)

(b) Hence evaluate $\sum_{r=35}^{50}(3r+5)$ (2)

Given that $\sum_{r=1}^{n}(3r+5) = 385$

(c) find the value of n. (3)

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9. Question 5 · part c · 3 marks

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Further practice · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 8 marks)

9. Question 5 · part c · 3 marks · Working and answer

Working and answer

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10. Question 3 · part c · 3 marks

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Further practice · Attempt part c only.

3 (a) Show that $\sum_{r=1}^{n}(5r-3)=\frac{n}{2}(5n-1)$ (3)

(b) Hence, or otherwise, evaluate $\sum_{r=31}^{60}(5r-3)$ (2)

Given that $\sum_{r=1}^{n}(5r-3)=3783$

(c) find the value of n (3)

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10. Question 3 · part c · 3 marks

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Further practice · Attempt part c only.

Question 3 continued

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10. Question 3 · part c · 3 marks · Working and answer

Working and answer

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11. Question 5 · part c · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

5 The sum of the first 10 terms of an arithmetic series A is $36k + 1$ where k is a constant.

The 6th term of A is $4k + 1$

(a) (i) Find an expression in terms of k for the common difference of A

(ii) Show that the first term of A is -8

(5)

Given that the 4th term of A is 7

(b) show that $k = 4$

(2)

The sum of the first n terms of A is S_n and the nth term of A is U_n

(c) Find the value of n such that $S_n = 5U_{n+10} + 105$

(4)

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11. Question 5 · part c · 4 marks

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Question 5 continued

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11. Question 5 · part c · 4 marks

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Question 5 continued

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11. Question 5 · part c · 4 marks

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Question 5 continued

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(Total for Question 5 is 11 marks)

11. Question 5 · part c · 4 marks · Working and answer

Working and answer

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